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Phase aberration compensation via a self-supervised sparse constraint network in digital holographic microscopy

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ABSTRACT

Phase aberration compensation is of great significance for the quantitative phase imaging in digital holographic microscopy. Least-square-based fitting algorithms are the most common options except for the double exposure method. However, they either require perturbation hypothesis, image segmentation, or an extra object-free hologram, leading to additional efforts for aberration elimination. In addition, supervised learning-based networks are mostly applied to create object-free masks automatically, but suffer from laborious label preparation and limited generalization capability. In this study, we propose a self-supervised sparse constraint network (SSCNet) for aberration compensation without the perturbation hypothesis, as well as an additional label or mask. Zernike model enhancement and sparse constraints are introduced for fitting aberration with only a single measured hologram. Simulation and experiment demonstrate that our proposed SSCNet can accurately compensate the phase aberration without object-free masks, and is more stable than other algorithms. It can be generalized to other samples with high accuracy and little time requirement. Moreover, SSCNet has advantage in the adaptive compensation of dynamic aberration in continuous measurement. Finally, it is an effective and practical fitting algorithm for phase aberration compensation.

1. Introduction

As we all know, quantitative phase imaging (QPI) technology has demonstrated its value in metrology, life science, and medicine [1,2]. Digital holographic microscopy (DHM) is a powerful QPI technique and has wide applications in biological research, thermal effect visualization, and flow field measurement, owing to its non-invasive, label-free, high-resolution, and quantitative phase-contrast visualization [3–5]. Unfortunately, the distortions in DHM, such as the coherent noise and undesired phase aberration, seriously affect the comparative observation and quantitative analysis. One contributing factor for phase aberration is the angle between the object and reference beams in the off-axis DHM setup; another factor is the deployment of microscope objective (MO). In addition to the tilt and quadratic phase aberration, astigmatism, coma, and other high-order aberrations may exist due to the unideal construction and component mismatch in an optical system. The phase aberration compensation is greatly important to improve the performance of phase-contrast visualization and the accuracy of quantitative measurement.

In recent years, various methods have been developed to eliminate phase aberrations, and they are broadly categorized into two groups, namely, physical compensation and numerical fitting. The former usually introduces the telecentric configuration [6,7] or electrically tunable lenses [8,9] in the optical setup. Although effective, it requires additional optical components or configurations and is highly dependent on the system accuracy and stability [10]. By contrast, the numerical fitting method can accomplish aberration compensation without additional holograms [11,12] or devices. The rolling ball algorithm is usually used for the background calibration in biological imaging [13], but is not very suitable for the aberration compensation in quantitative phase measurement. It affects the phase height of object easily in compensation, and spends more effort on manual parameter tuning and data conversion. An aberration-free phase can be reconstructed by using the algorithms, including lateral shear [14], wavefront folding [15], hologram

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geometric transformation [16–18], and polynomial model fitting [19–37], to fit the aberration numerically. Apart from the spectral analysis [19–21], the least-squares (LS) fitting is the most widely used to calculate the coefficients with different aberration models, such as parabolic function [19,20], spherical surface [22], standard polynomials [23–25], or Zernike polynomials [26–31]. However, the conventional LS algorithm has to satisfy the perturbation hypothesis, which assumes that the object phase is a small perturbation compared with the aberration. The principal component analysis (PCA) is proposed to reduce the impact of object and accelerate the fitting process, but it cannot compensate the cross-term aberration [32–34]. Recently, Liu et al. proposed two methods for wrapped phase based on phase variation minimization (PVM) and total standard deviation minimization. However, both methods still require phase unwrapping essentially in the process and only verify their actual performance under perturbation hypothesis [27,28]. Ren et al. used the sparse optimization (SO) in compress sensing theory to replace the L_2-norm with L_1-norm and achieved higher accuracy than LS under perturbation hypothesis [29]. Ma et al. created an object-free mask from the hologram and utilized the patch-based inpainting algorithm to generate the background hologram for compensation [30]. Besides, the use of image segmentation technique to generate an object-free mask for LS fitting is a common and useful solution [31,35,36]. However, these algorithms often require professional tuning and may be time-consuming and susceptible. The deep learning technique has shown great potential in dealing with image-related tasks and also plays a role in phase aberration compensation nowadays [35–41]. Ma et al. and Nguyen et al. used the deep convolutional neural networks to implement the rapid image segmentation [35,36]. Chang et al. utilized U-Net to learn the mapping from a measured phase to a corrected phase [40]. Xiao et al. considered the aberration compensation as a regression task with aberration models, where the optimal coefficients act as the responses to the input measured phase [41]. Nevertheless, most deep learning methods are only used for image segmentation, but have no physical connection with aberration compensation. Moreover, all networks belong to the supervised learning, indicating that they suffer from the long-time training with abundant labeled data and the poor practicability caused by limited generalization capability.

In general, most LS fitting methods are trapped in the perturbation hypothesis and are inapplicable in practice, while the image segmentation causes the aberration compensation to become more complex and cumbersome. Deep neural networks are still limited to supervised learning and not deeply integrated with the compensation task. We propose a self-supervised sparse constraint network (SSCNet) for phase aberration compensation to solve these flaws. The Zernike model enhancement and sparse constraints are introduced together to limit the solution space and reduce the impact of object phase. Only one measured phase is sufficient for self-supervision, which is highly compatible with the actual measurement. Results of simulation and experiment indicate that SSCNet can perform aberration compensation with high precision and stability, but without the object-free mask as well as the perturbation hypothesis. Furthermore, it has advantages over double exposure in the adaptive compensation of dynamic aberration.

2. Principle

2.1. Zernike aberration model

Through recording and reconstruction stages of DHM, the complex amplitude of the light field on the reconstruction plane is

$$U(x,y) = A(x,y)\exp\left[i\varphi_{mea}(x,y)\right] = A(x,y)\exp\left[i\left(\varphi_{obj}+\varphi_a+\varphi_n\right)\right] \tag{1}$$

where $A(x, y)$ is the amplitude term, and φ_{mea} is the total measured phase after unwrapping, which contains the object phase φ_{obj}, aberration φ_a, and coherent noise φ_n. Assuming that the noise term is negligible, the approximate object phase $\bar{\varphi}_{obj}$ can be easily calculated as long as the optimal fitting phase $\bar{\varphi}_a$ is obtained by

$$\bar{\varphi}_{obj}(x,y) = \varphi_{mea}(x,y) - \bar{\varphi}_a(x,y) \approx \varphi_{obj}(x,y) \tag{2}$$

$$\bar{\varphi}_a(x,y) = \sum_{i=1}^{n} \bar{a}_i Z_i(x,y) \approx \varphi_a(x,y) \tag{3}$$

where $Z_i(x, y)$ is the i_{th} Zernike polynomial in the Cartesian coordinates, and $\bar{a}_i$ is the corresponding optimal coefficient. Here, the Zernike aberration model is used uniformly for comparison, and other aberration models, such as standard polynomials, are still applicable. According to Eq. (3), the phase aberration compensation task is transformed into a coefficient optimization problem. Moreover, Zernike aberration model plays an important role in limiting the fitting space of phase. The object phase has a specific shape different from the aberration, so that it is difficult to be fitted exactly by aberration model with finite orders. Thereby, the coefficients derived from optimization can mainly fit the aberration in a limited space.

2.2. Self-supervised sparse constraint network (SSCNet)

The coefficients can be calculated by various fitting algorithms, such as the LS fitting [26–31], the masked LS fitting (MLS) [35,36], the quasi-newton solver [27,28], and the deep neural networks [41]. Except for the "end-to-end" mode from φ_{mea} to $\bar{\varphi}_a$ or $\bar{\varphi}_{obj}$ directly, the networks with aberration model are effective. This approach can be regarded as a physical model enhancement because the aberration model actually restricts the solution space. The optimization process of these networks can be roughly divided into three groups, as shown in Fig. 1(a). Under the supervised mode, the Net1 and Net2 utilize the coefficient and corresponding aberration as labels, whereas the Net3 takes advantage of the measured phase as label. The first two approaches can obtain high fitting accuracy for compensation, but acquiring abundant instantaneous aberrations for training is costly. The Net3 is more practical, but its convergence is unsatisfactory. Here, we train three networks to show their optimization processes. The mean square error (MSE) is recalculated between the measured and corrected phases for unified comparison, and the loss curves are shown in Fig. 1(b). The loss curves of Net1 and Net2 converge to the desired status, but that of Net3 differs, leading to the accurate compensation by networks except Net3 in Fig. 1(c). In fact, Net3 is closer to the LS fitting in terms of optimization process and compensation results. Given that the object phase affects the fitting process, can we reduce its impact rather than creating an object-free mask to remove?

Inspired by the deep image prior idea, a number of researchers consider other untrained network approaches with physical model enhancement [42–46]. This mode skillfully realizes self-supervision with a few images for optimization, which is much suitable for the aberration compensation. From this point of view, we propose the SSCNet for phase aberration compensation. It also takes the measured phase as label, similar to Net3, but is implemented with custom loss functions. In Fig. 2(a), the input is set as an initial array instead of an image, such as the all-one array, because it can greatly reduce the parameters of network without convolution. The optimization of neural nodes eventually leads to an array $\{c_1, c_2, \ldots, c_n, c_{n+1}\}$, where n is the number of coefficients required by Zernike model, and c_{n+1} is the amplification factor. To fully utilize the nonlinear ability of the network, c_1 to c_n are activated by "Tanh()" function and limited to $[-1, 1]$, so that c_{n+1} is directly derived as the amplification factor. Once the output array is obtained, we can calculate the fitting phase φ_f and the corrected phase φ_c as the following:

$$\varphi_f(x,y) = \sum_{i=1}^{n} a_i Z_i(x,y) = \sum_{i=1}^{n} \left(c_i \cdot c_{n+1}\right) Z_i(x,y) \tag{4}$$

$$\varphi_c(x,y) = \varphi_{mea}(x,y) - \varphi_f(x,y) \tag{5}$$

In fact, the network benefits the compressed sensing strategy by changing its direct output from the phase matrix to the coefficient

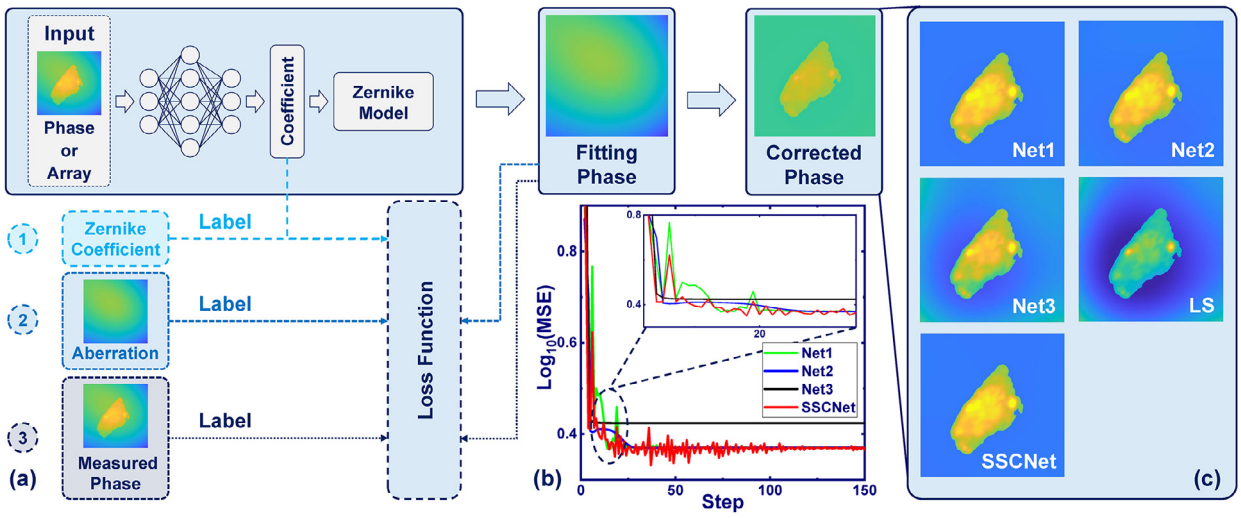


Fig. 1. Neural networks are applied to the coefficient optimization problem with Zernike aberration model. (a) Three networks with different labels for training: Net1 (coefficient), Net2 (aberration), and Net3 (measured phase). (b) Loss curves of all networks with MSE between the measured phase and fitting phase. (c) Corrected phases by Net1, Net2, Net3, LS fitting and our proposed SSCNet.

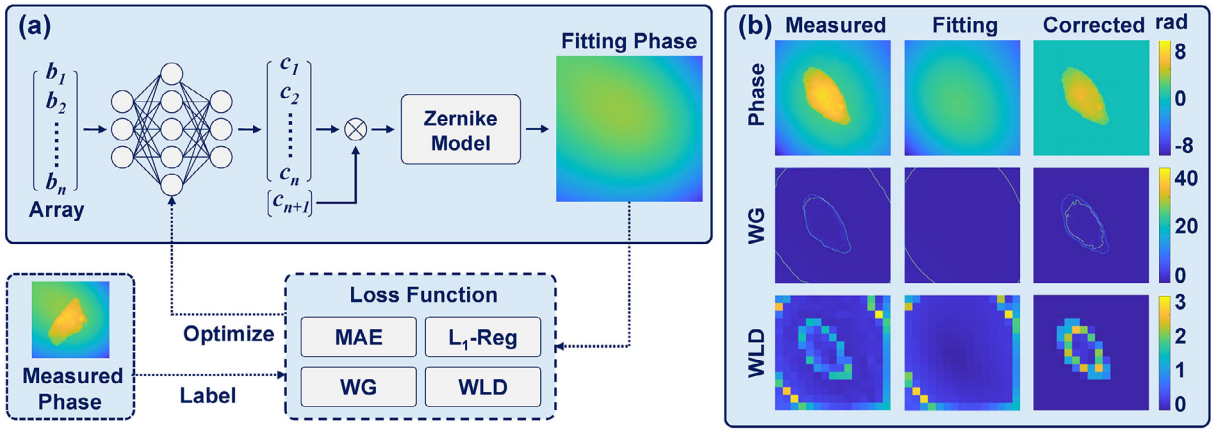


Fig. 2. Details of SSCNet. (a) Optimization process of SSCNet with custom loss functions, including L_1-norm (MAE), L_1-Reg, WG and WLD sparse constraints. (b) Gradient and local deviation matrices of wrapped phases, and the unwrapped phase matrices.

with aberration model enhancement. However, the object phase enables overfitting of the optimization process with the measured phase, as shown in Fig. 1. It is still an ill-posed inverse problem from φ_{mea} to $\bar{a}_i$ directly. Instead of obtaining a flat region using image segmentation in MLS, we turn to the sparse constraints in the loss function. Generally, the L_2-norm is a regression process to minimize the system energy, such as the LS fitting and Net3. Therefore, its solution is often non-zero and tends to vary uniformly, resulting in the poor performance in practice. We can strengthen the sparse constraint and obtain the solution with the least non-zero values using the generalized L_0-norm. At this time, each element in the solution is relatively opposed to each other during optimization, which is in line with the coefficients. The L_1-norm is always used in lieu of the L_0-norm for easier problem solving. Here, we perform the L_1-norm constraint for φ_c, which is equivalent to the mean absolute error (MAE) between φ_{mea} and φ_f. The out array $\{c_i\}$ is also constrained by the L_1-norm regularization (L_1-Reg). In addition, we transform φ_c into the wrapping domain and introduce the wrapped gradient (WG) and wrapped local deviation (WLD) sparse constraints to further reduce the impact of object phase. The WG sparse constraint calculates the gradient matrix from wrapped phase φ_{wc}, and then performs the L_1-norm. The WLD sparse constraint divides φ_{wc} into several local blocks with k^2 size, then calculates the absolute deviation values in each block, and finally executes L_1-norm for the block matrix. By doing so, the coefficient optimization problem becomes

$$f(a) = \frac{1}{w \cdot h}\left\{\lambda_1\|\varphi_c\|_1 + \lambda_2\left\|\{c_i\}\right\|_1 + \lambda_3\left\|G(\varphi_{wc})\right\|_1 + \lambda_4\left\|LD(\varphi_{wc})\right\|_1\right\} = Loss1 + Loss2 + Loss3 + Loss4 \tag{6}$$

$$G(\varphi) = \left|\varphi_{i+1,j} - \varphi_{i,j}\right| + \left|\varphi_{i,j+1} - \varphi_{i,j}\right| \tag{7}$$

$$LD(\varphi) = \sum_{i,j}\left|\varphi_{i,j} - \mathrm{Mean}\left[S_k(\varphi)\right]\right| \tag{8}$$

$$\left(\bar{a}_1, \bar{a}_2, \cdots, \bar{a}_n\right) = \arg\min f\left(a_1, a_2, \cdots, a_n\right) \tag{9}$$

where λ_1, λ_2, λ_3, are λ_4 are the scale factors, w and h are numbers of rows and columns, respectively, φ_c and φ_{wc} are the corrected phases before and after wrapping, respectively, G(•), LD(•), S_k(•), and Mean(•) are the functions for the gradient, local deviation, blocking, and averaging, respectively. We show an example in Fig. 2(b) to explain the role of wrapping operation. All phase matrices are non-sparse, but the sparsity is greatly improved after wrapping. The contours of WG and WLD matrices are highly similar to those of their phases, and the linear relationship of φ_{mea}=φ_f+φ_c is basically maintained. Nevertheless,

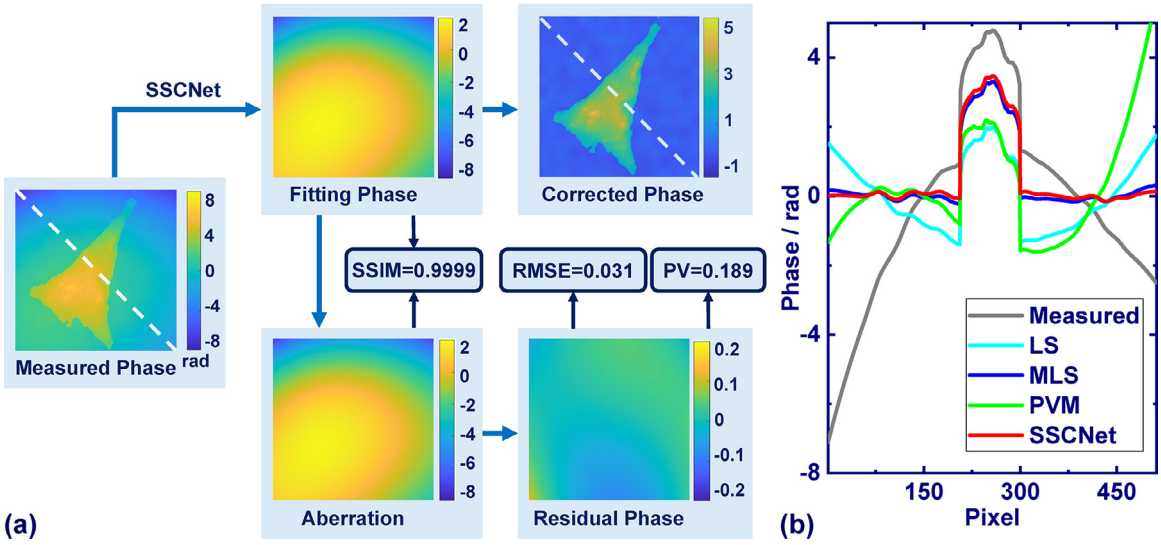


Fig. 3. A simulated cell for the procedure and compensation effect. (a) Correction procedure of SSCNet. RMSE: root mean square error; PV: peak to valley; SSIM: structural similarity. (b) Corresponding truncated lines of measured and corrected phases by different fitting algorithms. PVM: phase variation minimization; LS: least-squares fitting; MLS: masked least-squares fitting; SSCNet: our proposed method.

the impact of object phase is reduced tremendously in the two sparse matrices. With the loss function minimization, the algorithms can only fit the aberration and eliminate its contours, such that its final solution will be approximate to the ideal form. The phase shift leads to the jump of the contour. Thus, the L_1-norm and L_1-Reg constraints on the phase are utilized for balance. (Effects of different losses in Supplement 1.1) In Figs. 1(b) and (c), SSCNet with these losses can converge to the desired status and obtain the same compensation result as Net1 and Net2.

As for the long iteration time, we add the down-sampling and pre-generation strategies during the optimization process. The measured phase is down-sampled from larger size to 256^2 size for self-supervision, and the obtained coefficients are transformed into the fitting phase with the original size for compensation. However, the transformation process is still time-consuming because each term of Zernike polynomial needs to regenerate the phase matrix in each iteration step. We pre-generate the phases of all terms prior to optimization, so that only the coefficient multiplication is required. SSCNet with down-sampling and pre-generation significantly reduces the time from ~120 s to ~0.5 s for 200 steps. (Details and discussions in Supplement 1.2) As a result, SSCNet can handle the phase aberration compensation task immediately and accurately, but without perturbation hypothesis or image segmentation.

2.3. Simulation discussion

Compared with the aberration, the smaller the phase range and the proportion in the field of view (FOV) of object are, the closer the perturbation hypothesis will be. The special shapes of object phase may also have more influence on the fitting process due to the Zernike model. Hence, we discuss the actual performance of SSCNet in phase range, shape of object, and proportion in the FOV. The LS, MLS, and PVM algorithms are implemented in Python for comparison. Indexes, such as the structural similarity (SSIM) between the fitting phase and aberration, the root mean square error (RMSE), and the peak to valley (PV) of the residual phase, are implemented for evaluating the results. (Discussion about running time in Supplement 2) SSCNet is executed for 200 steps to ensure convergence. All algorithms are run on the computer with Intel Core i7–10,700 K CPU (3.8 GHz) and NVIDIA GeForce RTX 1080Ti GPU.

Here, we simulate a cell with aberration and noise, which does not satisfy the perturbation hypothesis to show the procedure and compensation effect in Fig. 3. As shown in Fig. 3(a), SSCNet can fit aberration and correct the phase perfectly. Fig. 3(b) shows the truncated lines of φ_{mea} and φ_c by various algorithms. The lines of SSCNet and MLS are the flattest, whereas LS and PVM fail to correct. (Comparison of coefficients in Supplement 2.3) This finding indicates that SSCNet can reduce the influence of object phase during fitting and obtain a compensation effect similar to MLS.

2.4. Phase range

We generate four data sets to verify the impact of phase range for algorithms. The 1st-3rd orders of Zernike polynomial are used for the aberration generation and fitting process of all algorithms because the 0th order only generates phase shift but does not affect the distribution of aberration. All simulated cells are controlled within similar phase range except for the aberrations. Sets 1, 2, and 3 are generated with 500 samples for simulating the small, medium, and large aberrations, respectively, whereas Set 4 is regenerated similarly to Set 2 but with noises. Noises are created according to the Gaussian distribution with 0.1 variance and 16^2 size and up-sampled to 512^2 size to add to the simulated cells and aberrations. Fig. 4 shows the comparison of compensation results by these algorithms, and the white lines in corrected phases are shown as the truncated lines right. The corrected phases of LS and PVM algorithms have large background phase fluctuation, indicating that they fail to fit the aberrations accurately. By contrast, those of MLS and SSCNet have significant correction effects. To examine the stability of algorithms, we define the correction success rate as: Success Rate = Count(RMSE $\leq$ 0.5 and PV $\leq$ 2.5 and SSIM $\geq$ 0.99) / N, where Count(•) is a function to count the number of qualified elements. The average indexes of four sets are listed in Table 1. From Sets 1 to 3 vertically, the samples become closer to the perturbation hypothesis, thereby increasing the SSIM and the success rate of LS. The fitting error also increases due to the increasing phase range, and the imperfect segmentation mask may lead to greater error. As for the PVM method, the success rate is relatively balanced but low on all three sets, indicating its non-stationarity on different samples. This condition may be caused by the inaccurate solving of quasi-Newton algorithm and the non-convexity of objective function. Contrastingly, SSCNet can fit the aberration perfectly and stably from Sets 1 to 3, although the training time increases slightly in Set 3. It is caused by more training steps (300 steps) given that the phase range of aberration is large. In Set 4, all algorithms maintain similar performance with the influence of noises

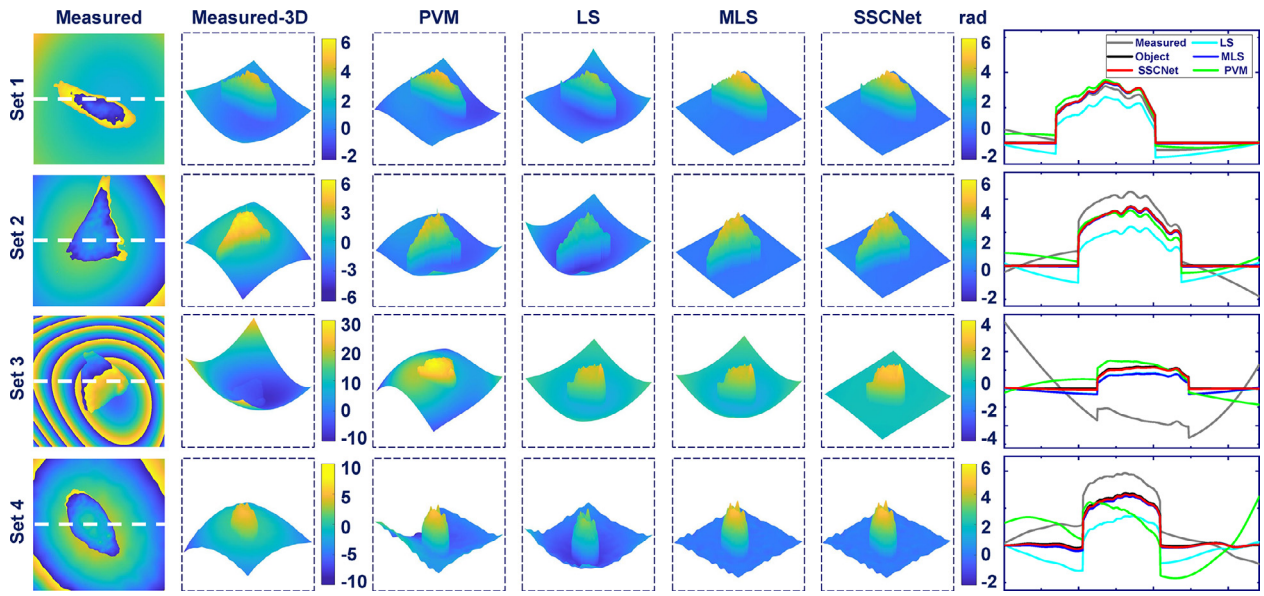


Fig. 4. Comparison of compensation results in four sets by algorithms. Sets 1, 2 and 3 are generated for simulating the small, medium and large aberrations, respectively. Set 4 is regenerated like Set 2 but with noises. White lines in corrected phases are the truncated lines right. PVM: phase variation minimization; LS: least-squares fitting; MLS: the masked least-squares fitting; SSCNet: our proposed method.

Table 1
Average indexes of four sets by algorithms. RMSE: root mean square error; PV: peak to valley; SSIM: structural similarity.

	Index	PVM	LS	MLS	SSCNet
Set 1	RMSE/PV	0.517/3.498	0.782 /3.486	0.042/0.191	**0.012/0.105**
	SSIM	0.9655	0.9178	0.9995	**1.0000**
	Success rate	15.0%	0.0%	99.4%	**100%**
	Time (s)	19.484	**0.479**	0.528	0.558
Set 2	RMSE/PV	0.830/4.986	0.764/3.386	0.070/0.343	**0.027/0.227**
	SSIM	0.9370	0.9308	0.9985	**0.9999**
	Success rate	10%	0.0%	95.6%	**100%**
	Time (s)	17.690	0.492	**0.490**	0.570
Set 3	RMSE/PV	1.626/9.021	0.787/3.490	0.381/1.927	**0.138/1.079**
	SSIM	0.8907	0.9630	0.9867	**0.9992**
	Success rate	12.4%	4.2%	60.2%	**99.2%**
	Time (s)	18.458	**0.508**	0.603	0.827
Set 4	RMSE/PV	0.778/4.484	0.775/3.466	0.081/0.422	**0.039/0.236**
	SSIM	0.9421	0.9286	0.9981	**0.9998**
	Success rate	16%	0.0%	95.8%	**100%**
	Time (s)	18.241	**0.458**	0.524	0.564

Table 2
Indexes of sample 1, 2 and 4 by algorithms.

	Index	PVM	LS	MLS	SSCNet
Sample 1	RMSE/PV	2.126/14.696	1.252/4.732	0.152/0.541	**0.018/0.189**
	SSIM	0.8208	0.8799	0.9973	**1.0000**
Sample 2	RMSE/PV	0.720/3.887	0.798/3.751	0.060/0.303	**0.023/0.179**
	SSIM	0.9772	0.9463	0.9996	**1.0000**
Sample 4	RMSE/PV	0.649/4.619	1.014/3.790	**0.012/0.058**	0.042/0.276
	SSIM	0.9765	0.9128	**1.0000**	0.9999

because the Zernike aberration model does not allow them to fit noises. Horizontally, SSCNet is better than other algorithms in all sets. This finding confirms the stability and accuracy of the proposed SSCNet in general.

2.5. Shape of object

Three object phases are generated as spherical, tilt, and cake structure with similar ranges, and the same aberration is added to form samples 1, 2 and 4. The proportion of objects in the FOV is ~30%. Fig. 5 shows their phases before and after compensation by the algorithms, and Table 2 lists their evaluation indexes. In vertical comparison, the corrected phase of sample 2 is slightly better than that of sample 4, whereas those of sample 1 are the worst in three samples by LS and PVM. It indicates that the shape of object affects the fitting process, and different shapes result in different fitting errors. The MLS method removes the object by mask and only uses the object-free region for fitting; thus, it obtains great and relatively stable compensation results. SSCNet also has excellent performance from visualization and indexes due to its sparse constraints on the phase and wrapping domain. In the right column of Fig. 5, the truncated lines of MLS and SSCNet can cover those of object accurately, whereas the others still have a huge gap. This result

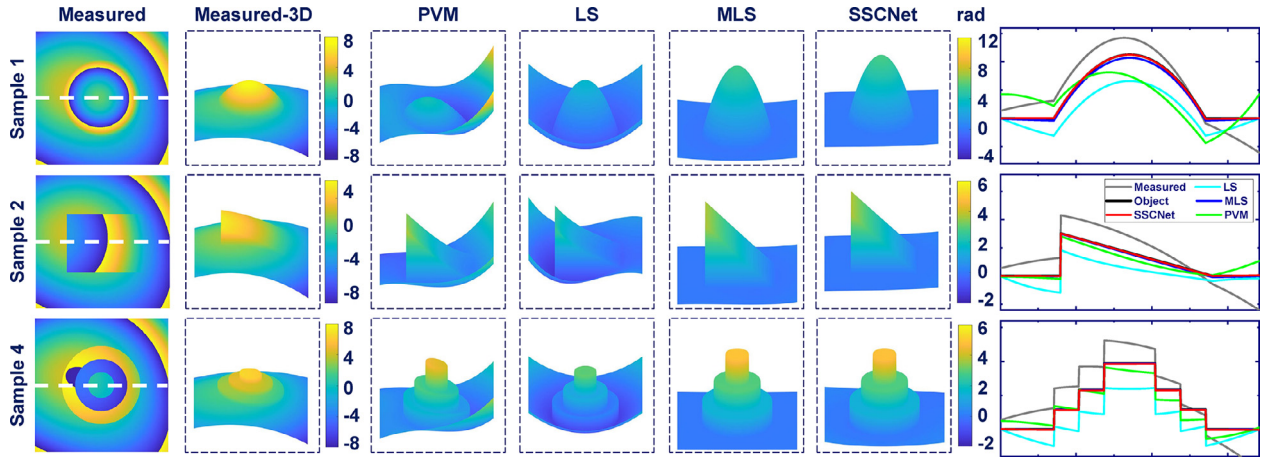


Fig. 5. Comparison of corrected phases with different objects by algorithms. White lines in corrected phases are the truncated lines right.

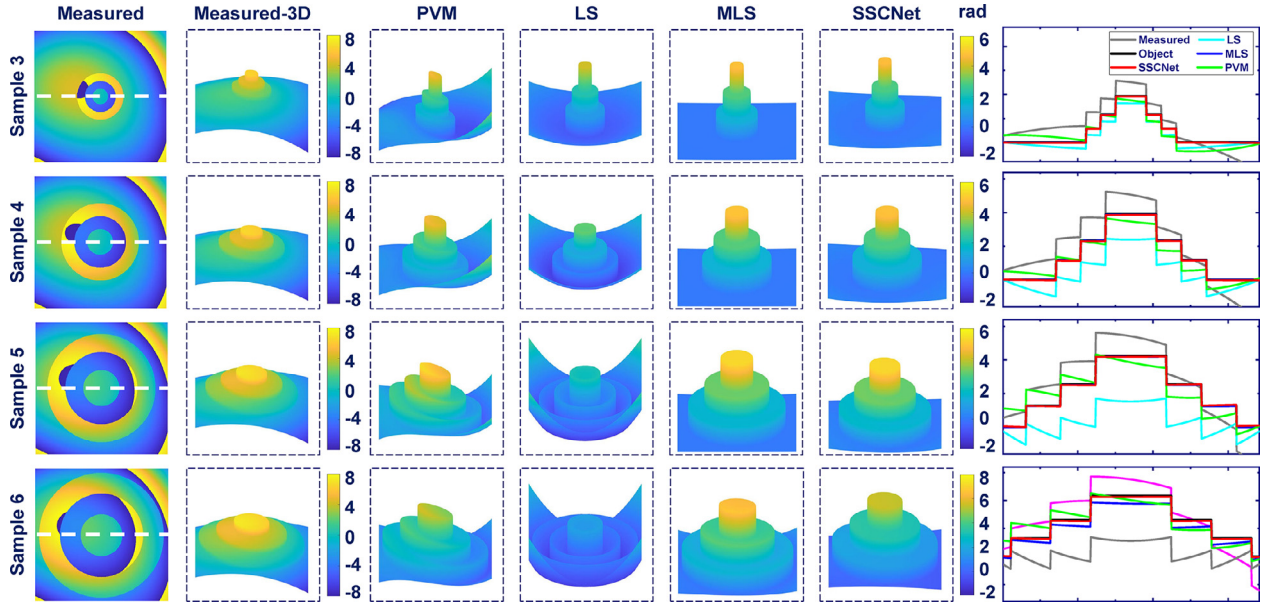


Fig. 6. Comparison of correction effect with different proportions in the FOV by algorithms. White lines in corrected phases are the truncated lines right.

Table 3
Indexes of sample 3, 4, 5 and 6 by algorithms.

	Index	PVM	LS	MLS	SSCNet
Sample 3	RMSE/PV	0.648/4.475	0.398/1.564	**0.011/0.044**	0.031/0.320
	SSIM	0.9683	0.9837	**1.0000**	0.9999
Sample 4	RMSE/PV	0.649/4.6190	1.014/3.790	**0.012/0.058**	0.042/0.276
	SSIM	0.9765	0.9128	**1.0000**	0.9999
Sample 5	RMSE/PV	0.770/3.904	1.869/6.185	**0.041/0.176**	0.085/0.651
	SSIM	0.9409	0.7756	**0.9998**	0.9994
Sample 6	RMSE/PV	0.705/2.725	2.276/6.668	0.429/1.549	**0.059/0.347**
	SSIM	0.9438	0.6992	0.9780	**0.9997**

confirms that SSCNet excludes the influence of object phase, although these special object phases have an impact on fitting.

2.6. Proportion in the FOV

Samples 3, 4, 5 and 6 are generated with the proportion of approximately 10%, 30%, 50%, and 70%, respectively, to explore the influence of proportions in the FOV. The aberration and phase ranges of objects are consistent. Fig. 6 and Table 3 represent the actual effect of four algorithms in these samples. Vertically, the compensation effect by LS and MLS decreases gradually from samples 3 to 6, and that by LS and PVM is unsatisfactory invariably in Fig. 6. In particular, the fitting accuracy of MLS decreases as the proportion increases in Table 3 because the higher proportion of objects leads to less object-free region for fitting after segmentation. In comparison, SSCNet is virtually unaffected and achieves high compensation accuracy on all samples.

As a result, the proposed SSCNet can achieve the accurate compensation of aberration with self-supervised learning, without the perturbation hypothesis or object-free masks. The accuracy and stability are verified with different objects, phase ranges, and even noises. It can avoid

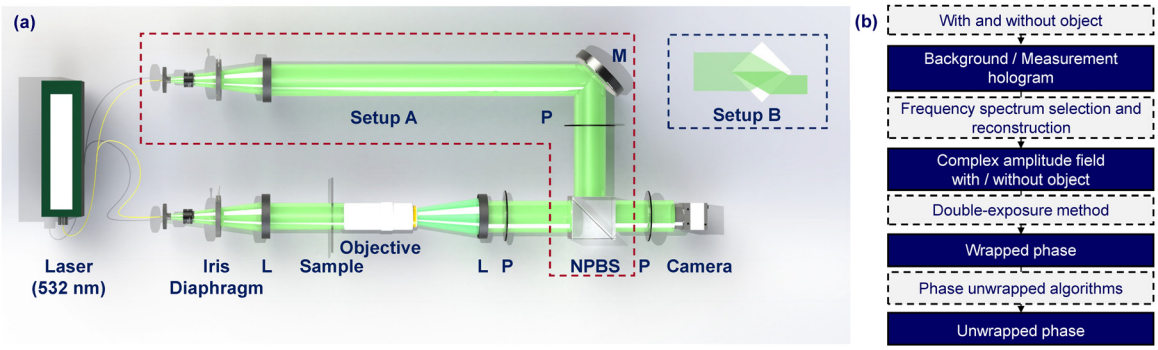


Fig. 7. (a) Experimental setup of an off-axis digital holographic microscope. L: lens, P: linear polarizer, M: mirror, NPBS: non-polarized beam splitter. (b) Phase reconstruction process by off-axis digital holography.

the impact of object phase and surpass the LS and PVM algorithms with special shapes of objects. Moreover, it is better than the three algorithms with the influence of proportion in the FOV.

3. Experimental setup and sample preparation

To acquire the actual data, we built an off-axis digital holographic microscope, as shown in Fig. 7(a). The laser source is Cobolt Samba TM 50 532 nm with a linewidth of <1 MHz. The green laser beam was collimated and incident to the sample. Through the microscope objective (MO), tube lens, linear polarizers, and non-polarized beam splitter (NPBS), the object beam was collected by a camera. The camera is Basler acA2040–90 μm with 2048 × 2048 pixels, and the pixel size is 5.5 × 5.5 μm^2. The reference beam was collimated and reflected to the camera for interference. By replacing setup A (red dotted lines) with setup B (blue dotted lines) in Fig. 7(a), the common-path setup was built by a non-polarized beam splitter (NPBS) placed at an angle of 45°. Three MOs (Mitutoyo M Plan Apo 20 ×, 50 × and 100 ×) were used for measurement. After getting the off-axis holograms, all phases were reconstructed according to the process [4,22] in Fig. 7(b). The double exposure method was used for compensating the aberration physically. At last, the unwrapping measured phases with aberration and corresponding object phases without aberration were reconstructed for verifying the performance of algorithms later.

Here we prepared six static samples and two dynamic samples for obversion. The Focus Star in Benchmark Quantitative Phase Target with refractive index 1.52 is the first static sample, which was captured with 20 × MO. The Cells 1, 2, 3 and the Microlens 1 were observed with 50 × MO, while the Microlens 2 was collected with 100 × MO. The Cells 1 and 3 are HT22 cells after overnight culturing. All cells were seeded in 3 cm dish (Nest), cultured in DMEM medium (BioInd) supplemented with 100 U/ml penicillin, 0.1 mg/ml streptomycin (BioInd) and 10% fetal bovine serum (Every Green, Tianhang) in a humidified atmosphere containing 5% CO2 at 37 ˚C. The Cell 2 is the dead animal cell specimen made by Suzhou Shenying Optics, and the Microlens array has the maximum height of 5 μm and refractive index of 1.458, produced by Opton. The two dynamic samples are the dividing osteoblasts and the changing ultrasound field. The acousto-optic crystal for ultrasound field is provided by CASTECH INC. The osteoblasts were captured by 20 × MO, while the changing ultrasound field was measured without MO.

The spare optimization (SO) [29] method is involved for comparison, due to its similar sparse constraints and good performance. However, we cannot directly determine the orders of Zernike model for fitting, thus, we fit the actual data by SSCNet, LS, and MLS with different orders (Details and discussions in Supplement 3). Results show that the 0th-3rd orders of Zernike model are sufficient for aberration compensation, indicating that low-order aberrations are still the main components in our experiment. As for the noise and special samples for over fitting issue, more details are presented and discussed in Supplement 4.

4. Experimental verification

4.1. Single measured phase

We initially compare the performance of all algorithms in terms of a single measured phase. Six samples are measured by changing the experimental setup and replacing the MO. They contain variations in phase range of aberration, noise, shape of object, and proportion in the FOV, which can be used to corroborate conclusions in Section 3. The reference aberration is obtained by double exposure for indexes. The aberration and corresponding fitting phase are subtracted from their mean values to exclude the influence of 0th-order. The corrected phases are visually displayed in Fig. 8, and the indexes for fitting accuracy are listed in Table 4. The PVM algorithm performs poorly on all samples with high time-consumption, indicating that it is inapplicable in practice. The Focus Star and Cell 1 are close to the perturbation hypothesis but other samples are not from phase range and proportion in the FOV. Thereby, the LS algorithms can easily obtain good compensation effects, but fit the aberrations unacceptably on other samples. With the object-free masks from image segmentation, the MLS algorithm has good effect, as shown in Fig. 8 and Table 4. The comparison of MLS and LS proves the effectiveness of object-free masks, but the image segmentation algorithms often require parameter tuning, which consumes considerable time in addition to computing. The SO method achieves better performance than LS in most samples, but the running time is extremely long. SSCNet can achieve better performance than SO because its sparse constraints are not only on the phase, but also on the wrapping domain. Furthermore, its elapsed time for optimization is only approximately ~0.5 s, which is much less than that of other algorithms due to the improvement of down-sampling and pre-generation. Consequently, SSCNet proves its accuracy and time advantage for compensating aberrations with a single measure phase, both from the visual observation of corrected phase and the quantitative comparison of evaluation indexes.

4.2. Dynamic measurement

The dividing osteoblasts are continuously measured by the common-path DHM setup. The unwrapped phases are compensated by the double exposure and SSCNet for comparison. Their corrected phases are partially shown in Fig. 9(a), and the complete results are shown in Visualization 1. In comparison, SSCNet obtains stable and effective compensation, whereas the corrected results of double exposure still have slight fluctuations over time. During long-term measurement, the total aberration of system may change slightly due to the additional disturbances. At this time, the effect of double exposure method is imperfect

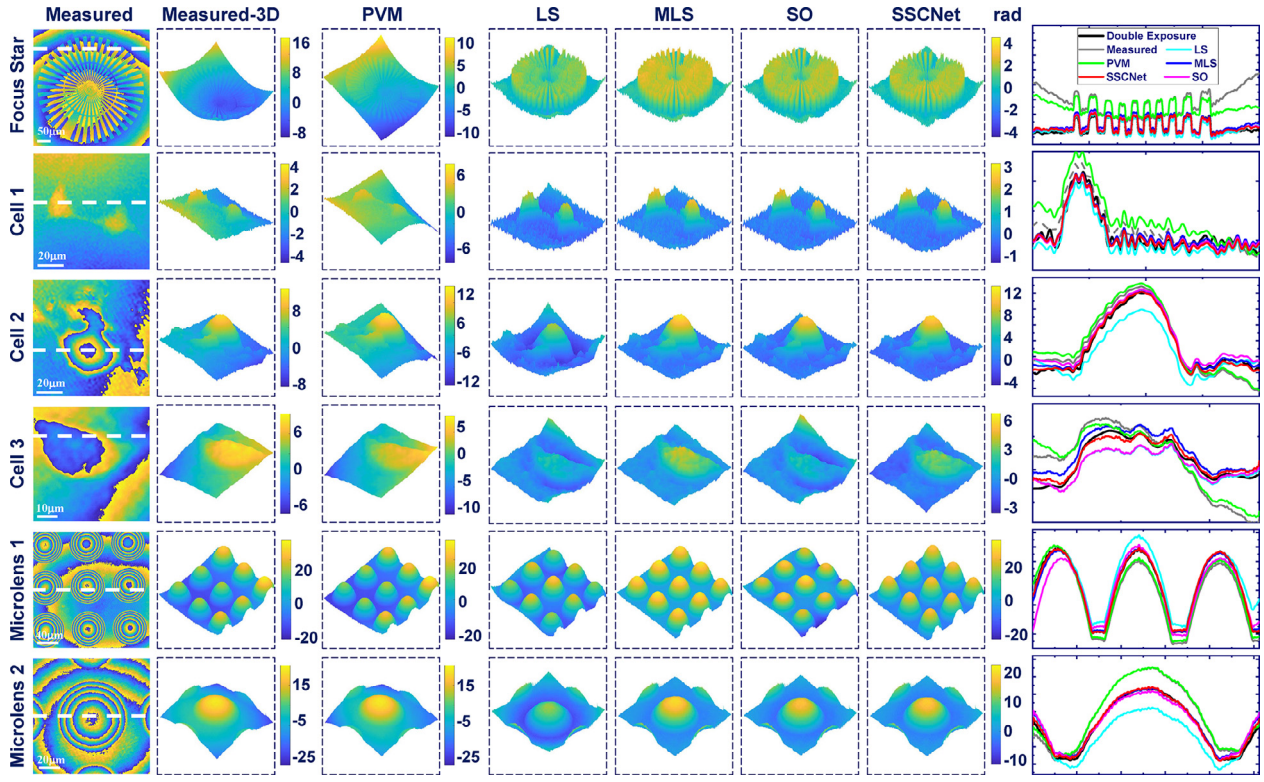


Fig. 8. Comparison of compensation effect by algorithms in experiment. Focus Star and Cell 1 both meet the perturbation hypothesis while other samples do not. These samples contain variations in phase range of aberration, noises, the shape of object and proportion in the FOV, which can be corroborated with the conclusions in simulation. White lines in corrected phases are the truncated lines right. PVM: phase variation minimization; LS: least-squares fitting; MLS: the masked least-squares fitting; SO: spare optimization; SSCNet: our proposed method.

Table 4
Indexes of all experimental samples by algorithms. RMSE: root mean square error; PV: peak to valley; SSIM: structural similarity.

	Index	PVM	LS	MLS	SO	SSCNet
Focus Star	RMSE/PV	3.362/19.178	0.370/3.013	**0.280/2.426**	0.311/2.669	0.327/2.806
(Size:1600^2)	SSIM	0.6195	0.9929	**0.9965**	0.9949	0.9942
	Time (s)	139.303	7.236	9.259	588.601	**0.563**
Cell 1	RMSE/PV	1.872/11.773	0.238/2.000	**0.178**/1.537	0.182/1.513	0.183/**1.426**
(800^2)	SSIM	0.7533	0.9947	**0.9970**	**0.9970**	0.9968
	Time (s)	38.840	1.218	1.420	111.679	**0.550**
Cell 2	RMSE/PV	2.658/16.739	1.556/10.553	0.386/2.906	0.550/4.347	**0.319/2.272**
(800^2)	SSIM	0.5944	0.8370	0.9850	0.9723	**0.9900**
	Time (s)	70.398	1.284	1.341	155.579	**0.555**
Cell 3	RMSE/PV	2.570/17.195	1.125/6.496	**0.407/2.647**	1.206/7.797	0.562/3.618
(512^2)	SSIM	0.6689	0.9193	**0.9860**	0.9110	0.9757
	Time (s)	15.465	**0.476**	0.563	43.411	0.554
Microlens 1	RMSE/PV	4.023/23.999	3.231/26.500	1.140/14.191	4.138/34.987	**0.432/8.221**
(2048^2)	SSIM	0.5057	0.7981	0.9707	0.7702	**0.9935**
	Time (s)	268.239	24.339	24.585	1101.599	**0.572**
Microlens 2	RMSE/PV	4.886/28.486	3.565/17.414	0.977/6.269	1.651/9.304	**0.441/5.490**
(2048^2)	SSIM	0.4636	0.8035	0.9707	0.9305	**0.9924**
	Time (s)	234.645	13.915	15.216	1610.292	**0.574**

due to its assumption of constant aberration on time. By contrast, these fitting algorithms have advantage in the adaptive compensation without the background hologram, and SSCNet has the highest accuracy among them without the perturbation hypothesis.

Another example is the continuous measurement of an acousto-optic crystal by setup A in Fig. 7. The ultrasound field activated by ultrasonic transducer is regarded as the object for measurement, which is of great significance for guiding the design and improving the diffraction efficiency. However, the phase of ultrasound field is always drowned in the total aberrations, including the system aberration, the phase of temperature change in crystal, and the phase of superimposed wave field due to the frequency fluctuation. The compensation results by double exposure and SSCNet are shown in Fig. 9(b), and more results are shown in Visualization 2. Although the two methods cannot compensate the wave field phase, SSCNet can fit the phase of temperature change adaptively, and has flatter and more stable results than the double exposure. As for the time consumption, SSCNet still requires ~0.5 s with 200 steps for compensation, which can be further compressed by the pre-training model. In general, SSCNet has evident advantage of adaptive compensation in the measurement with dynamic aberration compared with the double exposure.

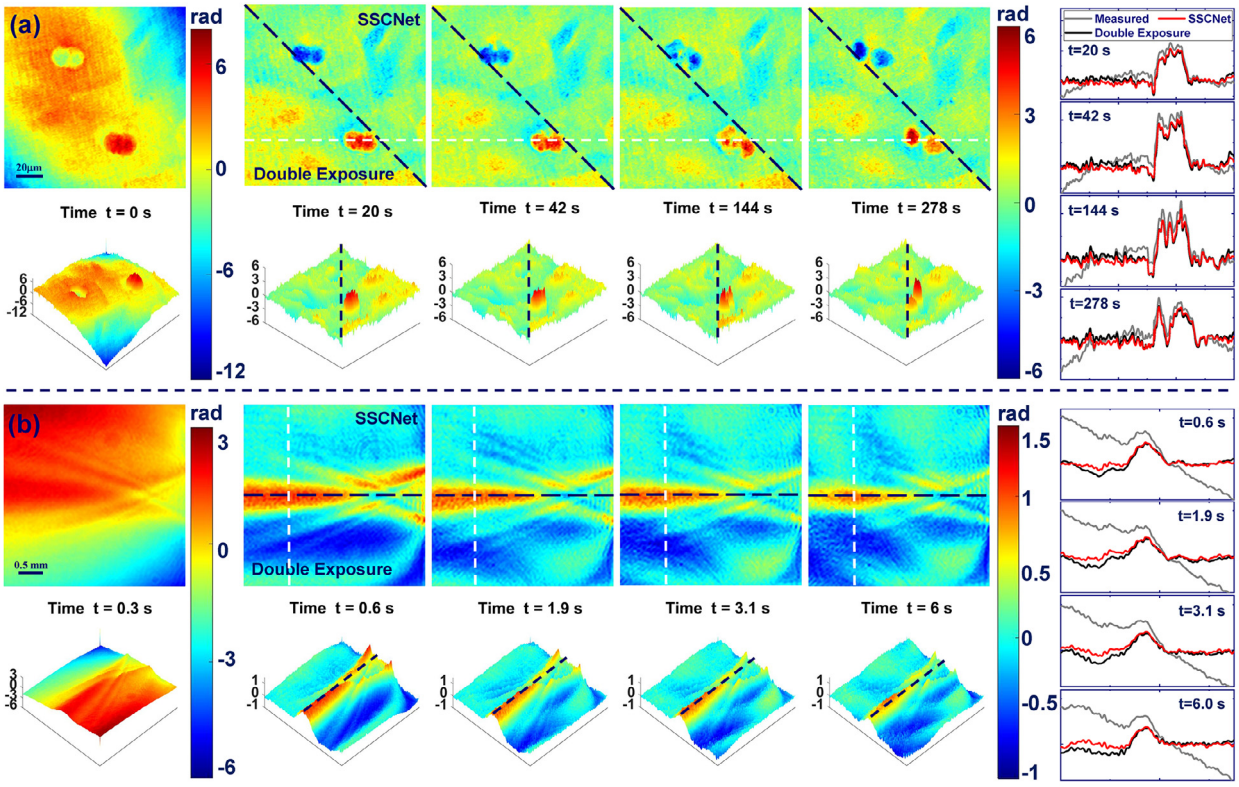


Fig. 9. Corrected phases of continuous measurement by double exposure and SSCNet in experiment. (a) Dividing osteoblasts measured by the common-path DHM setup with 20 × MO. (b) Ultrasound field in an acousto-optic crystal measured by off-axis setup without MO. The white lines in corrected phases are the truncated lines right.

5. Conclusions

In this paper, we propose a self-supervised sparse constraint network for phase aberration compensation in DHM. The proposed SSCNet combines the sparse constraints with Zernike model enhancement and only requires a single measured phase for self-supervision. Simulated discussion and experimental verification show that SSCNet can achieve accurate and stable compensation without an object-free mask, regardless of whether the perturbation hypothesis holds. Besides, we also discuss the influence of phase range of aberration, shape of object, and proportion of object in the FOV and SSCNet can surpass all conventional algorithms in all these respects. In the experiment, SSCNet also has excellent and stable compensation performance compared with other fitting algorithms, and it is better than the double exposure method in the adaptive compensation of dynamic aberration. On balance, our proposed SSCNet is effective and practical for phase aberration compensation not only in digital holography, but also in interferometry, adaptive optics, and so on.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ju Tang: Investigation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Jiawei Zhang:** Formal analysis, Methodology, Writing – review & editing. **Sijia Zhang:** Formal analysis, Writing – review & editing. **Shan Mao:** Writing – review & editing. **Zhenbo Ren:** Writing – review & editing, Supervision. **Jianglei Di:** Writing – review & editing, Supervision. **Jianlin Zhao:** Writing – review & editing, Supervision.

Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.optlaseng.2023.107671.

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